

## What is Machine Learning?

Machine Learning (ML) is a branch of artificial intelligence that enables computer systems to learn patterns from data and improve their performance on a task without being explicitly programmed for every rule. Instead of hard-coding instructions, an ML model is trained on examples and learns to make predictions or decisions based on that data.

## Why Machine Learning Matters

Traditional software follows fixed rules written by a programmer. Machine learning flips this: the system is given data and outcomes, and it works out the rules itself. This makes ML useful for problems that are too complex or constantly changing to hand-code, such as recognizing images, understanding language, detecting fraud, or recommending products.

## Types of Machine Learning

- **Supervised Learning** — the model learns from labeled data (input paired with the correct output), such as predicting house prices from historical sales data.
- **Unsupervised Learning** — the model finds patterns or groupings in unlabeled data, such as clustering customers by purchasing behavior.
- **Reinforcement Learning** — the model learns by taking actions in an environment and receiving rewards or penalties, such as an agent learning to play a game.

## Common Challenges in Machine Learning

- **Overfitting** — the model memorizes training data instead of learning general patterns, so it performs poorly on new data.
- **Underfitting** — the model is too simple to capture the underlying patterns in the data.
- **Data quality issues** — biased, incomplete, or noisy data leads to unreliable predictions.
- **Lack of interpretability** — some models (like deep neural networks) are difficult to explain, making it hard to understand why a prediction was made.

## How Machine Learning Relates to LLMs and RAG

Large Language Models are a specific application of machine learning, trained on massive text datasets to generate human-like language. As covered earlier, RAG is a technique used on top of LLMs to ground their responses in current, authoritative information rather than relying solely on what was learned during training.